



ESG investing

(Environmental, social, governance)

Michael Kwok



Motivation

- **ESG**

Invest in responsible and sustainable companies

- **Investigate:**

1. ESG vs non-ESG
2. ESG = higher returns?



Existing literature

- **Do ESG Exclusions Have an Effect on Portfolio Risk and Diversification?**
 - Yes but very limited effect
- **When Do Green Stocks Outperform Brown Stocks?**
 - Mainly during times when carbon emissions are not strongly factored into the implied equity risk premium
 - The question is not whether to overweight green stocks, but when.
- **Rethinking Corporate Priorities: The Surge OF ESG Investing And Greenwashing Risks**
 - ESG sacrifices profits? But attracts more investment?
- **An analysis of focusing on ESG investment**
 - Lack of standardized ESG data and consensus on what constitutes ESG factors





About

2004

- “ESG” is first mentioned by United Nations

2025

- \$30+ trillion AUM
- 130+ ESG ETF's:
 - ESGU, ESGV, EFIV, ...

Issues driving [Morningstar](#) / [Sustainalytics](#) ESG Risk Ratings^[54]

Category	Issue	Contribution to ESG Risk Rating
Environmental 43.3%	Carbon - Own Operations	19.2%
	Resource Use	10.3%
	Emissions, Effluents and Waste	7.1%
	Environmental and Social Impact of Products and Services	6.7%
Social 34.1%	Human Rights	22.8%
	Occupational Health and Safety	7.5%
	Community Relations	3.8%
Governance 22.6%	Corporate Governance	11.9%
	Business Ethics	6.7%
	Human Capital	4.0%

Challenges

Need: standardized/universal metrics/ratings

Subjective: some factors are measured more qualitatively than quantitatively

“Greenwashing”: companies can make misleading claims

Conflicting interests: quarterly financials vs long-term responsibility



1. ESG vs non-ESG

EFIV

SPDR® S&P 500® ESG ETF

Conceived in 2020 | \$1.48 billion AUM | ~300 holdings (60% of S&P 500)

“**Moderate**” ESG screening

**STATE
STREET**
GLOBAL
ADVISORS

Excludes companies that:

- Do not act in accordance with the **United Nations Global Compact principles**
 - (Regarding: **human rights, labor, environment, anti-corruption**)
- Are involved in **small arms, tobacco, thermal coal, controversial weapons, military contracting and oil sands**

Sector	SPY weight	EFIV weight	Difference
Information Technology	29.45%	31.85%	2.40%
Financials	14.59%	15.80%	1.21%
Health Care	11.01%	11.53%	0.52%
Consumer Discretionary	10.22%	7.66%	-2.56%
Communication Services	9.22%	7.49%	-1.73%
Industrials	8.60%	7.86%	-0.74%
Consumer Staples	6.50%	7.42%	0.92%
Energy	3.40%	4.11%	0.71%
Utilities	2.66%	1.35%	-1.31%
Real Estate	2.32%	2.57%	0.25%
Materials	2.03%	2.35%	0.32%

EFIV screening

ESG : (40% of S&P 500):

- Amazon, Domino's Pizza, Doordash, Ford, Johnson + Johnson, Netflix, Progressive, Southwest Airlines, Texas Instruments, Verizon, Workday...



ESG : (60% of S&P 500):

- Bank of America, Exxon Mobil, heavy AI-related companies (Microsoft?), ...



Bank of America



- CEO: "We are capitalists."
- North America's **largest financier** of **coal mining** (2023: \$1.5 billion)
- Coal mining:
 - Disrupts ecosystems
 - Destroys wildlife habitats
 - Diminishes biodiversity
 - Pollutes water (persists for decades)
- **Fossil fuels** are the **#1 contributor** to **greenhouse gas emissions**



SUSA



iShares MSCI USA ESG Select ETF

Conceived in 2005 | \$3 billion AUM | ~200 holdings (88% of its holdings are in S&P 500)

“**Stricter**” ESG screening; more mid-caps

Screens companies based on:

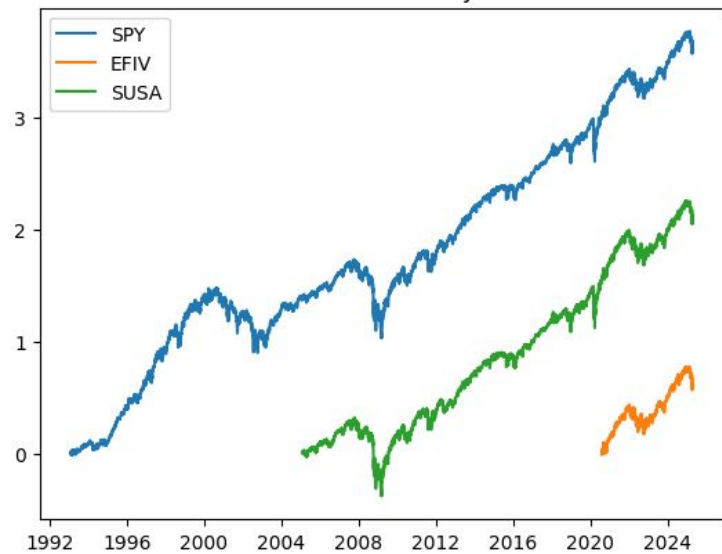
- Alcohol, civilian firearms, controversial weapons, conventional weapons, fossil fuel extraction, gambling, nuclear power, nuclear weapons, thermal coal power and tobacco.

Sustainability Characteristics

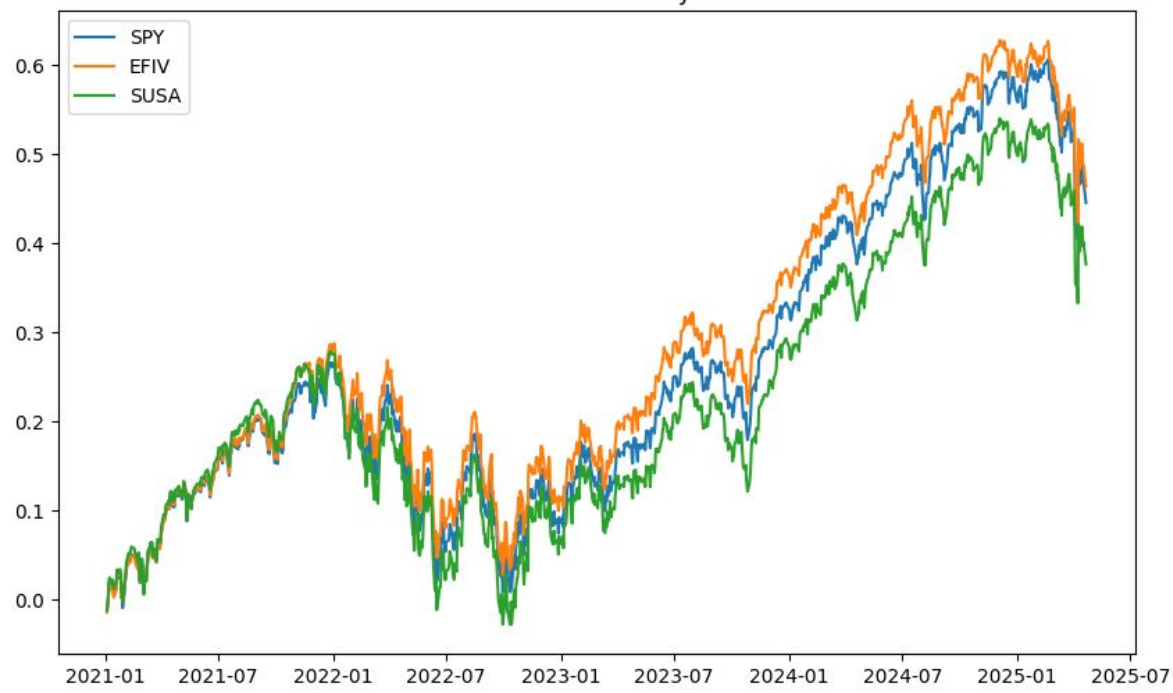
MSCI ESG Fund Rating (AAA-CCC)  as of Mar 21, 2025	AA
MSCI ESG Quality Score (0-10)  as of Mar 21, 2025	8.1
Fund Lipper Global Classification  as of Mar 21, 2025	Equity US
MSCI Weighted Average Carbon Intensity (Tons CO2E/\$M SALES)  as of Mar 21, 2025	57.97
MSCI Implied Temperature Rise (0-3.0+ °C)  as of Mar 21, 2025	> 2.0° - 2.5° C

2. ESG = higher returns?

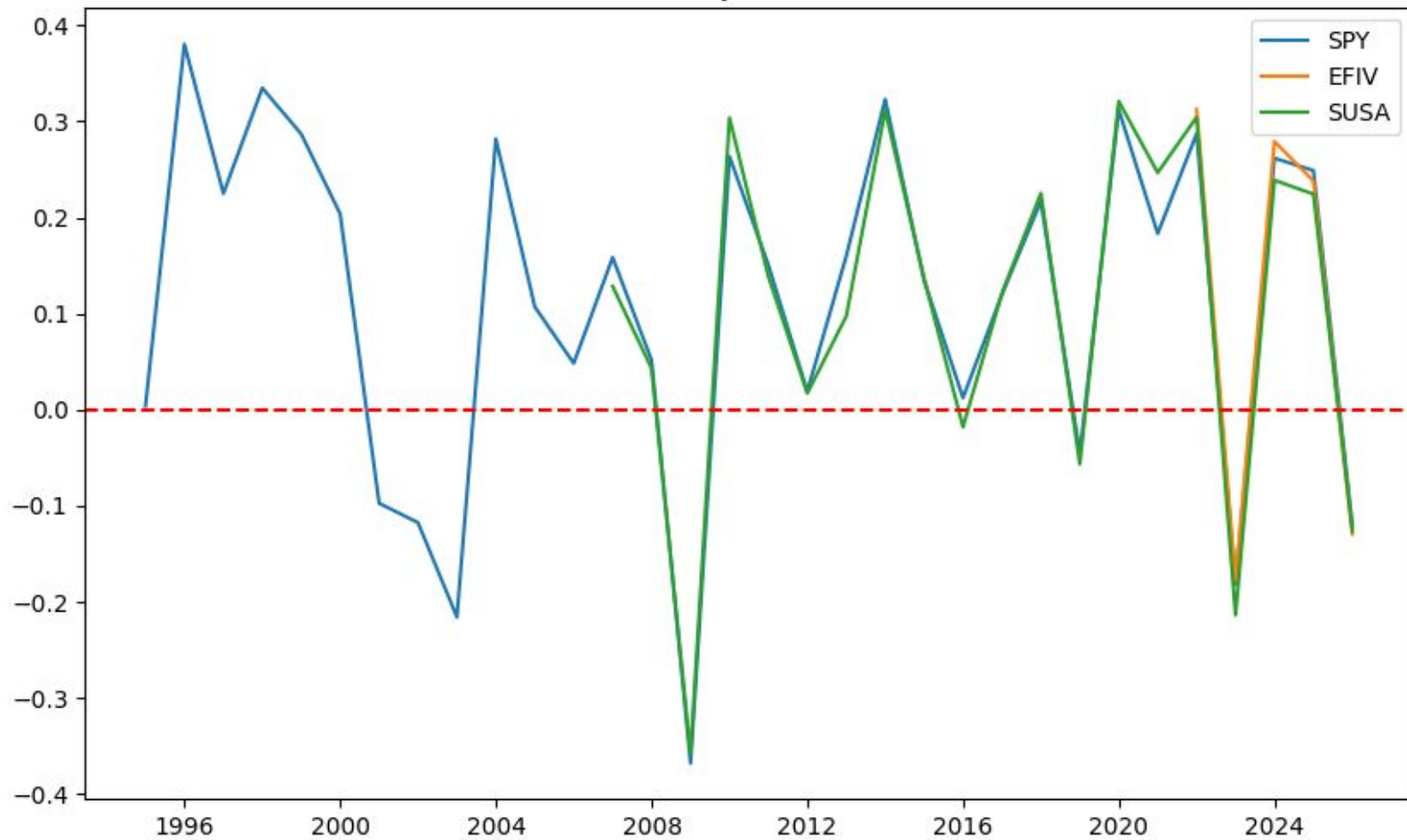
Cumulative sum of daily returns



Cumulative sum of daily returns

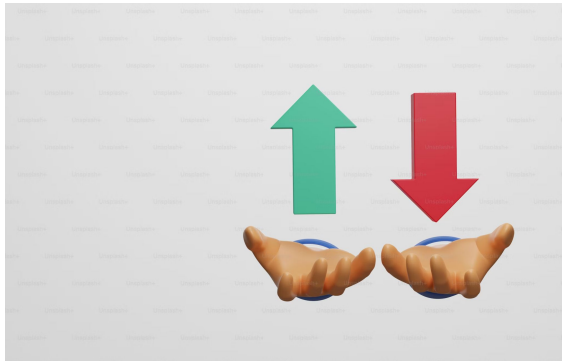


Yearly returns



Compare with SPY

	Annualized Ret	Sharpe	Daily corr SPY	Monthly corr SPY	Yearly corr SPY	# years	# companies
SPY	9.98%	0.4				23	500
EFIV	11.43%	0.27	0.99	0.996	0.998	5	300
SUSA	9.13%	0.34	0.96	0.988	0.99	20	200



Considerations

- Randomness
- Time horizon
- Sector-bias
- Correlation vs causation



Future Steps

- Additional topics to investigate with more time
 - **Various scoring methods:** robustness, standardization, ... (MSCI, Bloomberg, ...)
 - **Turnover analysis:** Inclusion and exclusion of specific firms in ESG index over time
 - **Examine other regions:** Europe and Asia



Related



FTSE4Good



FTSE4Good

- The FTSE4Good Index Series is a series of **ethical investment stock market indices** launched in 2001 by the FTSE Group[1] which reports on the performance of companies which demonstrate "strong Environmental, Social and Governance practices"

Principles for Responsible Investment

- The Principles for Responsible Investment Initiative (PRI) was established in 2005 by the United Nations Environment Programme Finance Initiative and the UN Global Compact as a **framework for improving the analysis of ESG** issues in the investment process and to aid companies in the exercise of responsible ownership practices

More ETFs

- XVV, SNPE, ESGU, ESGV, ...

Conclusion

- ESG vs non-ESG
 - Criteria is relatively complex and subjective
- ESG = higher returns?
 - Not necessarily, especially in the long-run
 - Many factors at play
- Better than nothing
 - Pressure companies to adopt responsible and sustainable practices



Appendix

<https://elicit.com/notebook/46c60fa3-85ff-44cb-8a80-58aabc2af004>

https://en.wikipedia.org/wiki/Environmental,_social,_and_governance

<https://www.ssga.com/us/en/individual/etfs/spdr-sp-500-esg-etf-efiv>

<https://www.ssga.com/us/en/intermediary/etfs/spdr-sp-500-etf-trust-spy>

<https://www.ishares.com/us/products/239692/ishares-msci-usa-esg-select-etf>

<https://unglobalcompact.org/what-is-gc/mission/principles>

<https://www.etf.com/topics/esg>

https://en.wikipedia.org/wiki/FTSE4Good_Index

https://bank.green/banks/bank_of_america

<https://environmentamerica.org/center/articles/how-coal-mining-harms-the-environment/>

Appendix (cont'd)

<https://www.reuters.com/markets/us/bank-america-ceo-says-we-are-capitalists-anti-esg-critics-gain-steam-2023-03-07/#:~:text=The%20lender%20outlined%20its%20ESG,in%20sustainable%20finance%20by%202030.&text=Lananh%20Nguyen%20is%20the%20U.S.,from%20the%20University%20of%20London.&text=Nupur%20Anand%20is%20a%20U.S.,an%20award%2Dwinning%20fiction%20writer.>

<https://www.theecoexperts.co.uk/news/top-7-most-polluting-industries>

https://ieefa.org/articles/esg-funds-continue-thrive-and-outperform-traditional-funds-across-equity-and-fixed-income?utm_source=chatgpt.com

<https://www.pm-research.com/content/pmrjesg/early/2025/04/22/jesg20251128>

<https://www.pm-research.com/content/pmrjesg/early/2025/04/10/jesg20251125>

<https://elicit.com/notebook/46c60fa3-85ff-44cb-8a80-58aabc2af004#1838299f5837986ffd9fa378e54b1706>

<https://elicit.com/notebook/46c60fa3-85ff-44cb-8a80-58aabc2af004#1838299f58138ac919ed4017bd8efe3c>